Project Report:
Analyzing Youtube Video Performance Using Machine Learning

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1. Introduction and Background

YouTube is one of the popular platforms for creating and sharing videos, with billions of views each day. Knowing what makes a video successful is important for content creators, advertisers, and YouTube itself. Videos that keep viewers watching and make money are more likely to be promoted by the platform.

However, predicting video performance is difficult due to many influencing factors—such as video length, upload time, viewer interactions, and monetization strategies. This project analyzes real YouTube data to identify the key factors that drive video engagement, retention, and revenue.

2. Research Questions

This project looks at three key measures of video success:

- **Average View Percentage** – What video characteristics influence the average view percentage of YouTube videos?
  Understanding which factors help retain viewers guides creators in optimizing content.
- **Watch time (hours)** – What factors influence the total watch time generated by YouTube videos?
  By identifying key factors that influence watch time, creators can optimize their content for better viewer retention and long-term audience growth.
- **Revenue per 1000 Views** - What factors influence the return of viewers and overall retention rates?
  Identifying its key drivers helps creators increase earnings and supports advertisers in selecting valuable content.

3. Dataset Description

To study this, we used a real YouTube dataset from Kaggle[1] with 364 videos and 70 features, including views, likes, comments, video length, revenue, and more. We grouped the data into categories like engagement, monetization, and viewer behavior to better understand what affects video performance. For detailed variable descriptions, refer to Appendix B.

4. Data Mining Method Description

This study used Multiple Linear Regression to analyze the relationships between YouTube performance metrics and variables. Linear regression was selected for its interpretability and ability to quantify predictor impact.

[1] https://www.kaggle.com/datasets/positivealexey/youtube-channel-performance-analytics

For Average View Percentage, multicollinearity was addressed using Variance Inflation Factor (VIF), followed by selecting significant predictors ($p < 0.05$) and applying Ridge Regression with GridSearchCV to reduce overfitting. A similar approach was used for Revenue per 1000 Views, excluding VIF analysis but including K-Fold Cross-Validation for model validation, with performance evaluated using R^2 and MSE. For Watch time (hours), predictor selection was guided by correlation analysis, and a Random Forest Regressor was applied to capture non-linear relationships and enhance accuracy. Residual analysis was conducted for all models to assess fit and identify potential biases.

This analytical approach ensured both interpretable insights and reliable predictions, supporting the study's goal of identifying key drivers of viewer engagement, retention, and monetization.

5. Exploratory Data Analysis, Cleaning and Modelling

5.1. Overview and Cleaning

The initial shape of the dataset 'df' was 364 rows and 70 columns. We took a few steps to standardize the dataset. The 'Video Publish Time' column required transformation, so we converted it to a datetime format. Then, the 'Day of Week' column contained categorical information, thus we encoded these categories into numerical values. The 'ID' column is just an identifier. Therefore, we removed it from the dataset.

We proceeded with a couple of overview and checking procedures. First, we checked for missing values, and no missing values were detected. We also examined the columns for a lack of information containing 0 values. Finally, we dropped these columns from the dataset. We obtained the updated dataset 'dfnew' which contains 364 rows and 50 columns. It contained columns with standardized data types. This cleaned dataset was ready for the following exploratory data analysis.

5.2. Exploratory Data Analysis

We took further steps to finalize the dataset. These steps ensured it was ready for further analysis. First, we visualized the numerical columns. Histograms helped us understand their spread. This gave us an initial overview.

Fig.1. Histograms of Numerical Columns in 'dffinal'

We focused on identifying unusual data. Boxplots were generated for all numerical columns. These plots helped us to see potential outliers visually. These columns 'Teaser Click-Through Rate (%)', 'Teaser Impressions', 'Teaser Clicks', 'Card Click-Through Rate (%)', 'Card Impressions', 'Card Clicks', 'Watched (Not Skipped) (%)', 'Feed Impressions 'seem less informative. We dropped the columns. This step created a new DataFrame named 'df2' which contains 364 rows and 42 columns.

Fig.2. Histograms of Numerical Columns in less information

Finally, we examined key numerical columns for outliers. The Interquartile Range (IQR) method helped identify extreme values. We dropped them. The finalized dataset is named

‘dffinal’ the shape was 335 rows and 42 columns. This step improved the reliability of further analysis.

5.2.1. EDA Overview

EDA started with an examination of correlations. We plotted a heatmap which showed the relationships between numerical columns in ‘dffinal’. It helped us find strong linear connections. Certain pairs of variables showed high correlation. Their absolute correlation was greater than 0.9. We extracted these strongly correlated pairs from the matrix.
Fig.3. Heatmap of Correlations Between Numerical Columns in ‘dffinal’

To understand these strong relationships, we plotted scatter plots based on the list of strongly correlated pairs. Each plot represented a strongly correlated pair. These plots helped us inspect the connection. We could see if the relationship was linear or non-linear, positive or negative, steady or sharply linear. This visual provided insight into how different variables relate to each other.
Fig.4. List of Strongly Correlated Pairs in ‘dffinal’
Fig.5. Scatter Plots of Strongly Correlated Pairs in ‘dffinal’

5.2.2. EDA Individual

We moved to analyze individual variables. Histograms provided insights into distributions. We visualized 'Video Duration' and 'Revenue per 1000 Views (USD)'.In terms of the plot, most of the video durations concentrated in the 500-700 second range.
Fig.6. Histogram of ‘Video Duration’

Scatter plots also showed connections. We analyzed 'Views' with 'Likes', 'New Comments', and 'Shares'. These are key engagement metrics. These metrics seemed to have a linear relationship.
Fig.7. Scatter Plot of ‘Views' with 'Likes', 'New Comments', and 'Shares'

Further, we investigated the correlation among engagement metrics. A heatmap displayed this. It showed how metrics like views and likes relate. The heatmap showed that views, likes, shares, and new comments show strong positive correlations, while average view duration and percentage have weak negative correlations with these metrics, and video thumbnail CTR exhibits a moderate positive correlation.
Fig.8. Heatmap of Correlations Among Engagement Metrics

6. Modeling and Findings

6.1. Average View Percentage

To examine the factors influencing Average View Percentage, a Multiple Linear Regression model was developed. Statistically significant predictors ($p < 0.05$) were selected, and multicollinearity was addressed using Variance Inflation Factor (VIF) analysis. The final model included ten predictors and demonstrated strong explanatory power, with an R^2 of 0.785 and an adjusted R^2 of 0.779. The overall model was statistically significant (F-statistic = 128.1, $p < 0.001$).

The data was split into 80% training and 20% testing sets. On the training set, the model achieved an R^2 of 0.81 and MSE of 14.53, while the test set yielded an R^2 of 0.66 and MSE of 26.84, indicating reasonable generalization.

To mitigate overfitting, Ridge Regression with cross-validation and hyperparameter tuning ($\alpha = 1$) was applied. This improved cross-validated performance to an R^2 of 0.73 and reduced MSE to 20.25, enhancing model robustness while retaining interpretability. This modeling workflow enabled both reliable prediction and clear identification of the key factors driving viewer retention.

Findings

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     Average View Percentage (%)   R-squared:                       0.785
Model:                                     OLS   Adj. R-squared:                  0.779
Method:                          Least Squares   F-statistic:                     128.1
Date:                         Fri, 09 May 2025   Prob (F-statistic):          9.81e-100
Time:                                 05:50:15   Log-Likelihood:                -916.28
No. Observations:                          325   AIC:                             1853.
Df Residuals:                              315   BIC:                             1890.
Df Model:                                    9
Covariance Type:                     nonrobust
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                      -1058.3626    286.338     -3.696      0.000   -1621.739    -494.986
Video Duration                -0.0446      0.002    -24.536      0.000      -0.048      -0.041
Year                           0.5703      0.144      3.968      0.000       0.288       0.853
Like Rate (%)                 -0.4841      0.146     -3.312      0.001      -0.772      -0.197
Dislikes                      -0.0057      0.003     -2.119      0.035      -0.011      -0.000
Views per Playlist Start      -0.3087      0.138     -2.231      0.026      -0.581      -0.036
Playlist Views                -0.0052      0.001     -6.682      0.000      -0.007      -0.004
Playlist Watch Time (hours)    0.0450      0.012      3.913      0.000       0.022       0.068
Average View Duration          0.1026      0.006     16.659      0.000       0.091       0.115
Impressions                 1.051e-06   3.65e-07      2.882      0.004    3.33e-07    1.77e-06
==============================================================================
Omnibus:                      158.471   Durbin-Watson:                   1.789
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2254.728
Skew:                           1.631   Prob(JB):                         0.00
Kurtosis:                      15.485   Cond. No.                     1.57e+09
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.57e+09. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The strongest positive predictor was Average View Duration with a coefficient of +0.1026 and $p < 0.001$, confirming that longer session watch time significantly boosts overall view percentage. Playlist Watch Time (hours) had a smaller but still significant positive coefficient of +0.0450 ($p < 0.001$), showing that time spent engaging with playlists is linked to higher retention. The Year of publication also had a significant positive coefficient of +0.5703 ($p < 0.001$), suggesting that more recently published videos are watched more fully.

In contrast, Video Duration had a strong negative effect with a coefficient of −0.0446 ($p < 0.001$), meaning longer videos are associated with lower percentages of completion. Surprisingly, Like Rate (%) showed a negative coefficient of −0.4841 ($p = 0.001$), suggesting that highly liked videos might be shorter and not necessarily watched in full. Dislikes had a smaller negative effect (−0.0057, $p = 0.035$), and Views per Playlist Start and Playlist Views had coefficients of −0.3087 and −0.0052 respectively (both $p < 0.05$), reinforcing the idea that passive playlist consumption reduces average viewing percentage.

Impressions had a small but positive effect (+1.051e-06, $p = 0.004$), indicating that increased exposure slightly boosts viewer retention.

Practical Implications

This analysis highlights that to improve average view percentage, creators should focus on producing concise, well-structured videos that keep viewers engaged from start to finish. Longer videos tend to reduce viewer retention, while higher average view duration strongly correlates with better performance. Relying on passive playlist views or vanity metrics like likes and subscriber counts may not yield meaningful improvements. Instead, content should be designed to maximize active watch time through storytelling and clear pacing.

These findings provide a valuable, data-driven foundation for improving content strategy. For creators, it means crafting videos that hold attention rather than just attract clicks. For advertisers and brands, targeting content with high average view duration presents better engagement opportunities and more effective ad placements—ultimately leading to stronger returns on investment (ROI).

6.2. Watch Time (Hours)

After analyzing our data to report our findings to the question "What factors influence the retention of viewers?", we were able to discover multiple components that are statistically significant in predicting what keeps viewers engaged in the content. We ran an Ordinary Least Squares (OLS) regression, and the results showed the R-squared to be 0.978, meaning that around 97.5% of the variation in watch time is explained by our features. We found many variables that contribute to retention, with some of the most impactful being Video Duration, New Comments, Shares, Views, and Impressions.

To validate and enhance the robustness of our analysis, we also trained a Random Forest Regressor, which achieved an R-squared of 0.968 on the test data, confirming high predictive accuracy. Residual distribution analysis showed no significant bias, and an actual vs. predicted plot demonstrated a close alignment between predicted and real values. This modeling process allowed us to confidently identify the most influential factors for viewer retention and ensure reliable predictions despite multicollinearity among certain variables.

Findings

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     Watch Time (hours)   R-squared:                       0.976
Model:                            OLS   Adj. R-squared:                  0.974
Method:                 Least Squares   F-statistic:                     432.5
Date:                Thu, 08 May 2025   Prob (F-statistic):          2.74e-222
Time:                        19:52:52   Log-Likelihood:                -2799.6
No. Observations:                 325   AIC:                             5657.
Df Residuals:                     296   BIC:                             5767.
Df Model:                          28
Covariance Type:            nonrobust
==========================================================================================
                                     coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                           5.072e+05   1.42e+05      3.581      0.000    2.28e+05    7.86e+05
Video Duration                     1.9894      0.421      4.724      0.000       1.161       2.818
Days Since Publish                 1.7244      5.966      0.289      0.773     -10.017      13.465
Day                                5.4698      9.325      0.587      0.558     -12.882      23.822
Month                             -8.5352     26.275     -0.325      0.746     -60.245      43.175
Year                            -253.5537     71.443     -3.549      0.000    -394.154    -112.954
CPM (USD)                        145.6771    125.098      1.165      0.245    -100.517     391.872
YouTube Premium (USD)           -426.0368    330.302     -1.290      0.198   -1076.075     224.001
New Comments                       0.0927      0.801      0.116      0.908      -1.483       1.669
Shares                            -4.0952      0.854     -4.794      0.000      -5.776      -2.414
Like Rate (%)                      2.5453     57.969      0.044      0.965    -111.538     116.628
Dislikes                          -2.4622      1.493     -1.650      0.100      -5.399       0.475
Likes                              0.1348      0.106      1.272      0.204      -0.074       0.343
Unsubscribes                      -1.5374      1.309     -1.175      0.241      -4.113       1.038
New Subscribers                   -0.3896      0.623     -0.625      0.532      -1.616       0.837
End Screen Click-Through Rate (%)  30.5782     35.503      0.861      0.390     -39.293     100.449
End Screen Impressions            -0.0147      0.014     -1.032      0.303      -0.043       0.013
End Screen Clicks                 -0.0988      0.266     -0.372      0.710      -0.621       0.424
Views per Playlist Start         -61.8329     50.105     -1.234      0.218    -160.439      36.774
Playlist Views                    -0.2852      0.310     -0.919      0.359      -0.896       0.325
Playlist Watch Time (hours)       -1.1161      5.162     -0.216      0.829     -11.275       9.043
Clip Watch Time (hours)          739.6707   1034.730      0.715      0.475   -1296.690    2776.031
Clip Views                       -19.0113     15.687     -1.212      0.227     -49.884      11.861
YouTube Premium Watch Time (hours)  64.6319      3.764     17.172      0.000      57.225      72.039
YouTube Premium Views             -5.0065      0.415    -12.070      0.000      -5.823      -4.190
Average View Percentage (%)       60.6715     15.080      4.023      0.000      30.995      90.348
Views                              0.0747      0.003     24.285      0.000       0.069       0.081
Subscribers                        1.1479      0.771      1.488      0.138      -0.370       2.666
Impressions                        0.0006      0.000      2.197      0.029    6.22e-05       0.001
Video Thumbnail CTR (%)           14.1968     41.450      0.343      0.732     -67.378      95.771
==============================================================================
Omnibus:                      113.667   Durbin-Watson:                   1.847
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              984.396
Skew:                          -1.183   Prob(JB):                    1.74e-214
Kurtosis:                      11.191   Cond. No.                     1.39e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 2.68e-18. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

Video Duration has a coefficient of 1.9894 and a P-value of 0.000, showing a strong positive correlation which we can interpret that the longer videos are associated with more total watch time. Views has a coefficient of 0.0747 and a P-value of 0.000, suggesting that it is the strongest direct driver of retention, more views means there is more watch time. Impressions have a coefficient of .0006 and a P-value of 0.000, showing that more impressions usually boost retention and watch time. Shares have a coefficient of -4.0953 and a P-value of 0.000, showing a strong negative correlation, interestingly suggesting that videos that are shared more are usually short and not deeply watched. Variables such as Like Rate, End Screen Clicks, Playlist Watch Time (hours), Video Thumbnail CTR (%), New Subscribers all have high P-values, showing that they are not statistically significant in viewer retention and they do not meaningfully affect the outcome compared to other variables.

Practical Implications

The practical implications for content strategy is that creators should mainly focus on making longer, more engaging videos because it usually results in more watch time. More views and impressions lead to more retention and by boosting visibility through optimized titles, thumbnails, and search engine optimization (SEO), it can correlate with higher watch time. Shares don't always mean more engagement, because shared videos might be for virality but it does not keep people watching. Focusing on quality content that is able to sustain peoples interest throughout is more important. High click through rate can be misleading because a strong title or thumbnail might get users attention at first, but if the video does not deliver viewers will leave. Like rate and new subscribers aren't direct indicators of retention, although it is good it does not always reflect actual viewer engagement or time spent watching.

6.3. Revenue Per 1000 Views (USD)

To analyze the Revenue per 1000 Views (RPM), a systematic modeling approach was followed. Backward selection was first used to retain only the statistically significant features with p-values less than or equal to 0.05. A Multiple Linear Regression model was then trained on these selected features. Before modeling, all features were standardized using StandardScaler to ensure fair weighting and comparability. The model's performance was evaluated using an 80:20 train-test split, applying Mean Squared Error (MSE) and R^2 as the main metrics. To further assess the model's stability and generalizability, K-Fold Cross-Validation was conducted. Finally, Ridge Regression was implemented, and GridSearchCV was used to optimize the alpha regularization parameter for improved predictive performance.

Findings

```
Adjusted Model Summary (After Dropping Insignificant Features):
                            OLS Regression Results
==============================================================================
Dep. Variable:     Revenue per 1000 Views (USD)   R-squared:                       0.799
Model:                                      OLS   Adj. R-squared:                  0.794
Method:                           Least Squares   F-statistic:                     143.4
Date:                          Thu, 08 May 2025   Prob (F-statistic):           4.17e-84
Time:                                  02:30:35   Log-Likelihood:                 439.98
No. Observations:                           260   AIC:                            -864.0
Df Residuals:                               252   BIC:                            -835.5
Df Model:                                     7
Covariance Type:                      nonrobust
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                          0.0885      0.003     31.538      0.000       0.083       0.094
Year                           0.0515      0.004     13.590      0.000       0.044       0.059
Playback-Based CPM (USD)       0.0562      0.007      8.197      0.000       0.043       0.070
CPM (USD)                     -0.0397      0.007     -6.083      0.000      -0.053      -0.027
Ad Impressions                -0.0147      0.005     -2.702      0.007      -0.025      -0.004
YouTube Ads Revenue (USD)      0.0557      0.006      9.607      0.000       0.044       0.067
Likes                         -0.0370      0.003    -11.171      0.000      -0.044      -0.030
Playlist Watch Time (hours)   -0.0146      0.004     -3.812      0.000      -0.022      -0.007
==============================================================================
Omnibus:                       23.571   Durbin-Watson:                   2.128
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               75.278
Skew:                          -0.275   Prob(JB):                     4.50e-17
Kurtosis:                       5.578   Cond. No.                         5.21
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

The model explains 79.4% of the variance in Revenue per 1000 Views (Adjusted $R^2 = 0.794$), indicating strong predictive power. All predictors included in the final model are statistically significant ($p < 0.01$), suggesting their effects on RPM are unlikely due to random chance.

Among the positively associated predictors, Year (coefficient = +0.0515) indicates that newer content years are linked to higher RPM, Playback-Based CPM (USD) (coefficient= +0.0562) shows that higher advertiser payouts per monetized playback increase RPM, and YouTube Ads Revenue (USD) (coefficient = +0.0557) reveals that direct ad revenue is a strong contributor to RPM.

On the other hand, negatively associated predictors include CPM (USD) (coefficient = -0.0397), where surprisingly, higher, general CPM is linked to lower RPM; Ad Impressions (coefficient= -0.0147), which slightly reduces RPM; Likes (coefficient = -0.0370), where higher engagement via likes may not translate into monetized views; and Playlist Watch Time (hours) (coefficient = -0.0146), which is modestly associated with lower RPM, possibly due to content being consumed outside individual monetized views.

Practical Implications

The model shows that monetization quality is a primary driver of Revenue per 1000 Views (RPM), with Playback-Based CPM and YouTube Ads Revenue having strong, positive impacts. Interestingly, some commonly valued engagement metrics—such as CPM (USD), Likes, and Playlist Watch Time—are negatively associated with RPM, indicating that high engagement doesn't always lead to higher earnings. With an Adjusted R^2 of 0.794, the model demonstrates strong reliability and predictive power. To improve RPM, creators should prioritize enhancing Playback-Based CPM, increasing Ads Revenue, and publishing more recent content, as these factors contribute most significantly to higher monetized returns.

Appendix A: Figures

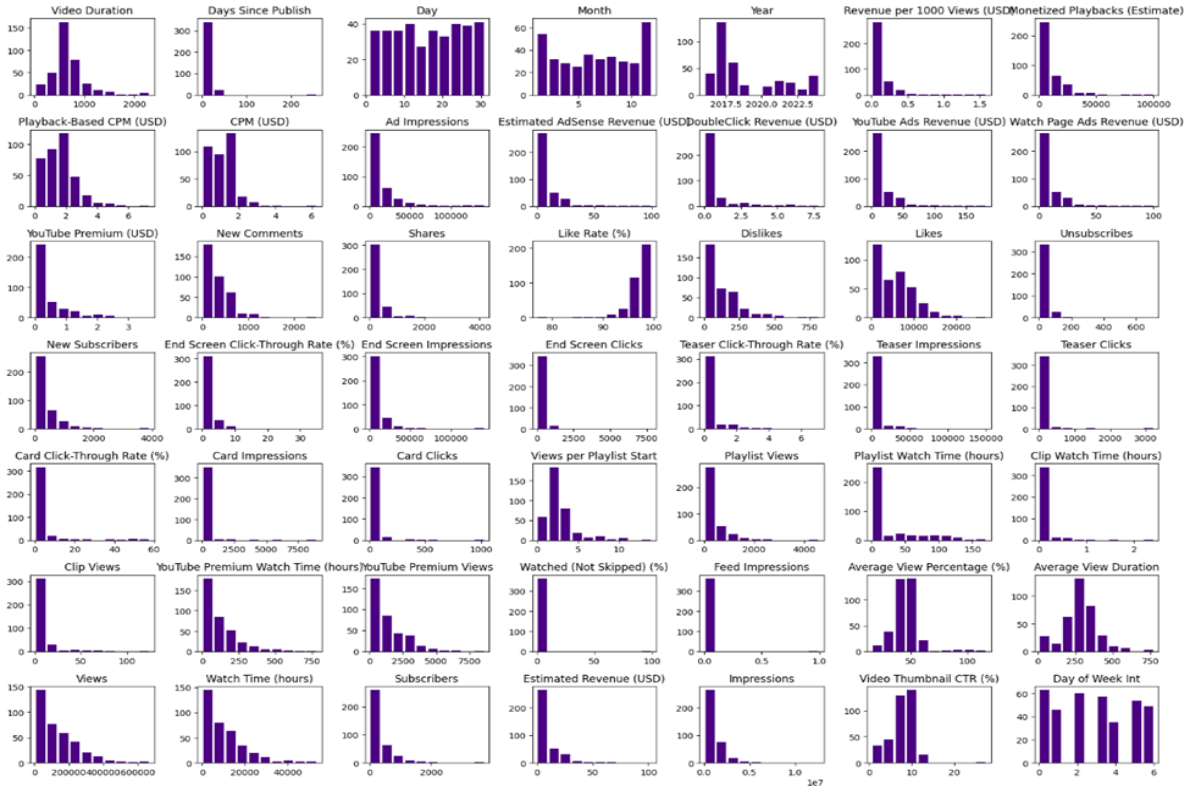

Fig.1. Histograms of Numerical Columns in ‘dffinal’

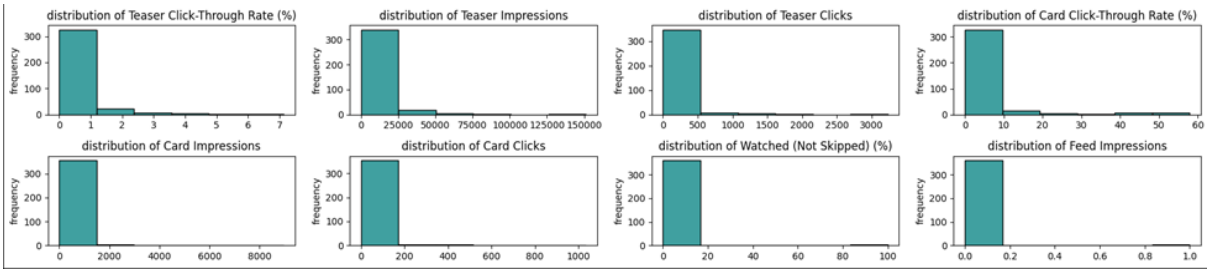

Fig.2. Histograms of Numerical Columns in less information

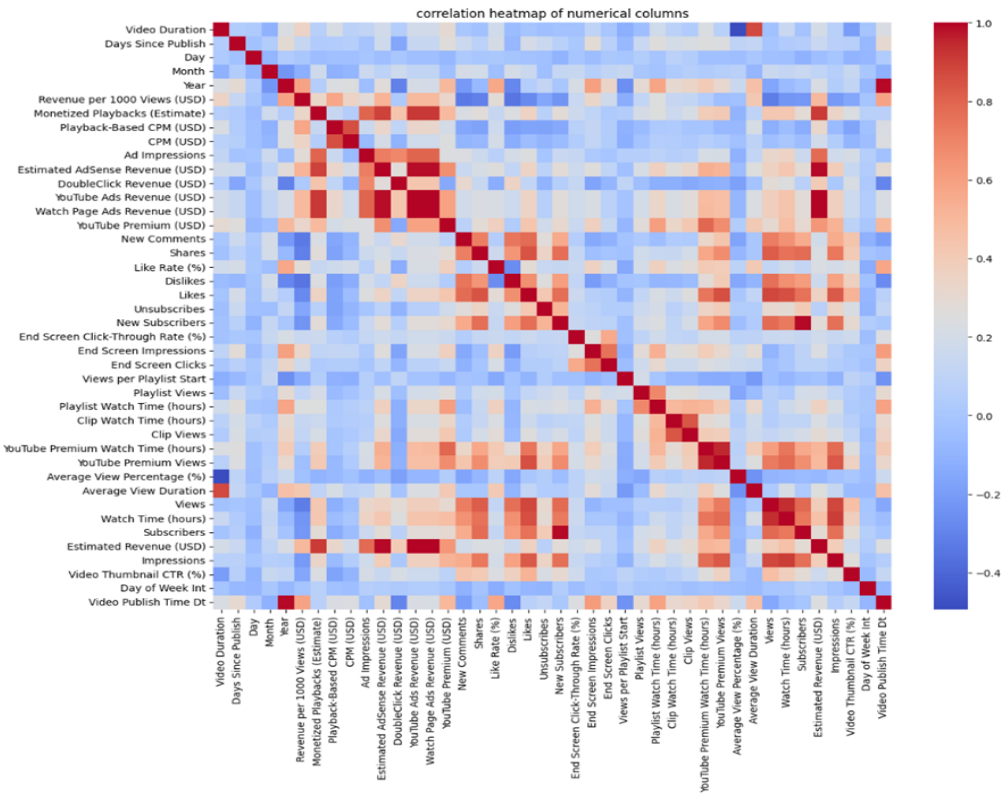

Fig.3.Heatmap of Correlations Between Numerical Columns in ‘dffinal’

	Variable 1	Variable 2	Correlation
0	Monetized Playbacks (Estimate)	YouTube Ads Revenue (USD)	0.912912
1	Monetized Playbacks (Estimate)	Watch Page Ads Revenue (USD)	0.912978
2	Monetized Playbacks (Estimate)	Estimated Revenue (USD)	0.908697
3	Estimated AdSense Revenue (USD)	YouTube Ads Revenue (USD)	0.986704
4	Estimated AdSense Revenue (USD)	Watch Page Ads Revenue (USD)	0.986754
5	Estimated AdSense Revenue (USD)	Estimated Revenue (USD)	0.989912
6	YouTube Ads Revenue (USD)	Watch Page Ads Revenue (USD)	0.999938
7	YouTube Ads Revenue (USD)	Estimated Revenue (USD)	0.998853
8	Watch Page Ads Revenue (USD)	Estimated Revenue (USD)	0.998908
9	New Subscribers	Subscribers	0.991359
10	YouTube Premium Watch Time (hours)	YouTube Premium Views	0.948599
11	Views	Watch Time (hours)	0.940379
12	Watch Time (hours)	Impressions	0.901090

Fig.4.List of Strongly Correlated Pairs in ‘dffinal’

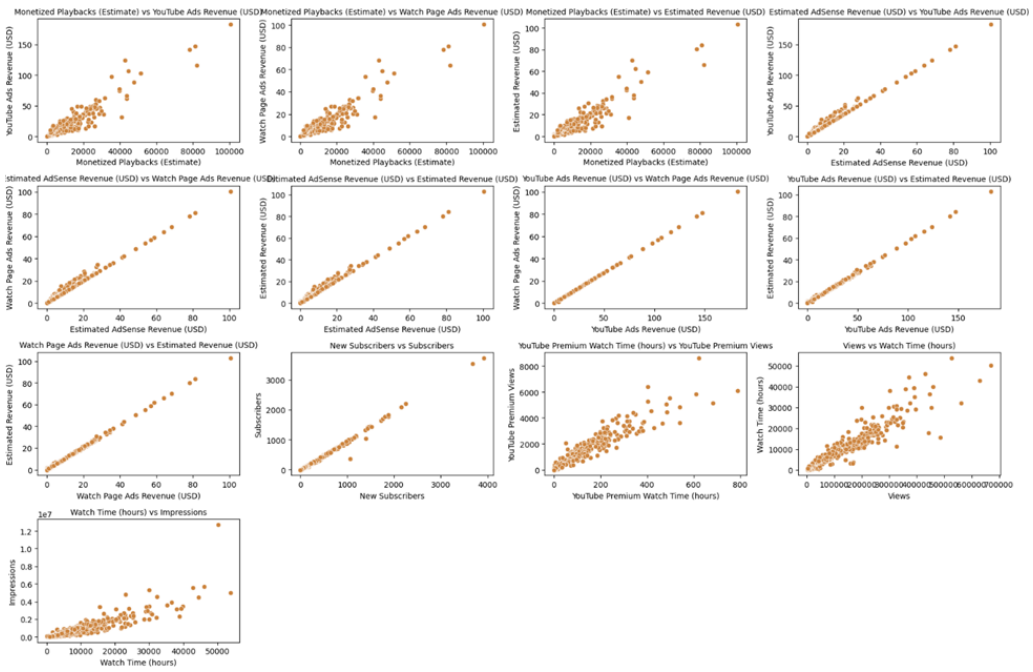

Fig.5.Scatter Plots of Strongly Correlated Pairs in ‘dffinal’

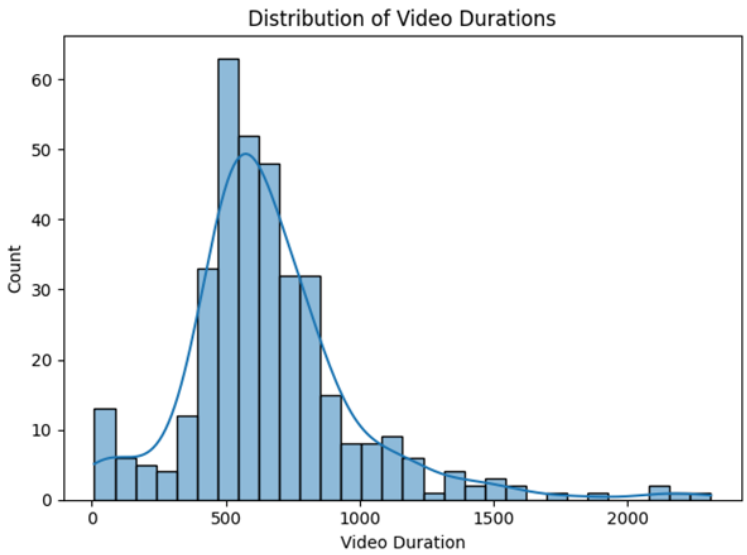

Fig.6.Histogram of ‘Video Duration’

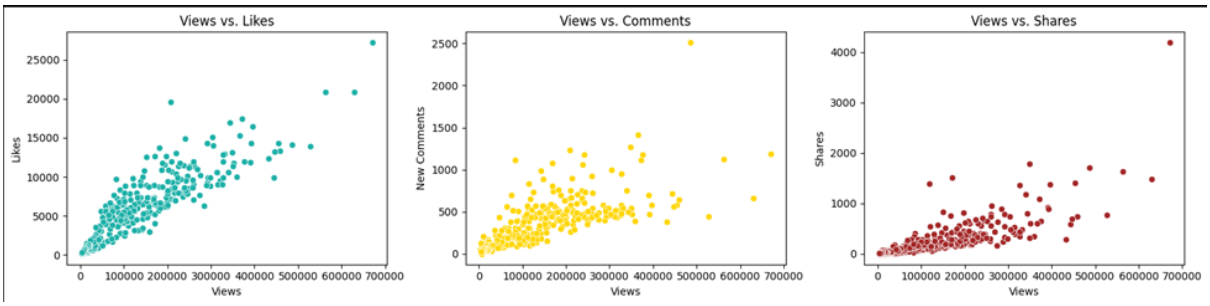

Fig.7.Scatter Plot of ‘Views' with 'Likes', 'New Comments', and 'Shares'

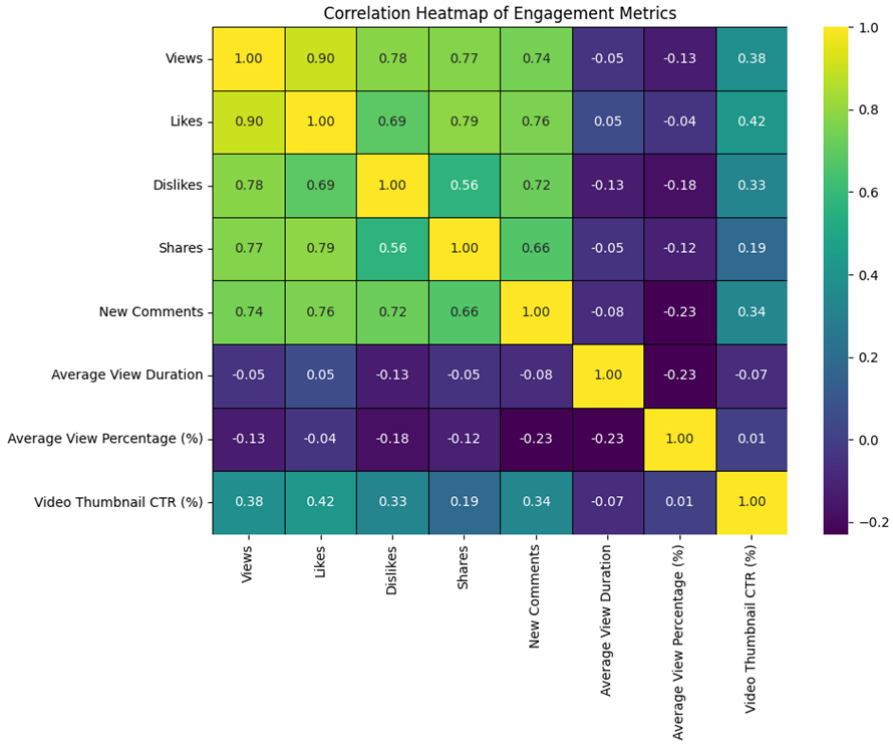

Fig.8. Heatmap of Correlations Among Engagement Metrics

Appendix B: Columns in Dataset

Video information:

1. **ID** – Unique identifier for each video entry.
2. **Video Duration** – Length of the video in seconds.
3. **Video Publish Time** – Date and time when the video was published.

Publishing date metrics:

4. **Days Since Publish** – Number of days since the video was published.
5. **Day** – Day of the month when the video was published.
6. **Month** – Month when the video was published.
7. **Year** – Year when the video was published.
8. **Day of Week** – Day of the week when the video was published.

Revenue metrics:

9. **Revenue per 1000 Views (USD)** – Estimated revenue generated per 1000 views.
10. **Monetized Playbacks (Estimate)** – Estimated number of monetized playbacks.
11. **Playback-Based CPM (USD)** – Cost per thousand monetized playbacks.
12. **CPM (USD)** – Cost per thousand impressions.
13. **Ad Impressions** – Number of times ads were shown on the video.
14. **Estimated AdSense Revenue (USD)** – Estimated revenue earned from AdSense.
15. **DoubleClick Revenue (USD)** – Revenue generated from DoubleClick ads.
16. **YouTube Ads Revenue (USD)** – Total revenue earned from YouTube ads.
17. **Watch Page Ads Revenue (USD)** – Revenue from ads displayed on the watch page.
18. **YouTube Premium (USD)** – Revenue generated from YouTube Premium viewers.
19. **Transaction Revenue (USD)** – Revenue generated from transactions.
20. **Transactions** – Number of transactions made through the video.
21. **Revenue from Transactions (USD)** – Revenue earned from transactions in USD.
22. **Returned Items (USD)** – Revenue lost due to returned items.
23. **Unconfirmed Commissions (USD)** – Pending commissions that are yet to be confirmed.
24. **Approved Commissions (USD)** – Approved commissions earned from sales.
25. **Orders** – Total number of orders placed through the video.
26. **Total Sales Volume (USD)** – Total revenue generated from sales.
27. **Estimated Revenue (USD)** – Total estimated revenue from all sources.

Engagement metrics:

28. **Reactions** – Total number of reactions received on the video.
29. **Chat Messages Count** – Total number of chat messages during live streams.
30. **Reminders Set** – Number of reminders set for a scheduled video.
31. **Subscribers from Posts** – Number of subscribers gained from posts.
32. **New Comments** – Number of new comments on the video.
33. **Shares** – Total number of times the video was shared.
34. **Like Rate (%)** – Percentage of likes relative to views.
35. **Dislikes** – Total number of dislikes received.
36. **Likes** – Total number of likes received.
37. **Unsubscribes** – Number of users who unsubscribed after watching the video.
38. **New Subscribers** – Number of new subscribers gained.

Remix and stream metrics:

39. **Stream Hours** – Total hours streamed.
40. **Remix Views** – Number of views from remixed versions of the video.
41. **Remix Count** – Total number of remixes created from the video.

Viewer and watch metrics:

42. **Returning Viewers** – Total number of returning viewers.
43. **New Viewers** – Total number of new viewers.
44. **Average Views per User** – Average number of views per user.
45. **Unique Viewers** – Total number of unique viewers.
46. **Views** – Total number of views on the video.

47. **Watch Time (hours)** – Total watch time in hours.
48. **Subscribers** – Number of subscribers gained from the video.

Watch time and retention metrics:

49. **YouTube Premium Watch Time (hours)** – Total watch time from YouTube Premium viewers.
50. **YouTube Premium Views** – Total number of views from YouTube Premium viewers.
51. **Watched (Not Skipped) (%)** – Percentage of ads watched without being skipped.
52. **Feed Impressions** – Number of times the video appeared in users' feeds.
53. **Average View Percentage (%)** – Percentage of the video watched on average.
54. **Average View Duration** – Average duration viewers spent watching the video.

Card and end screen metrics:

55. **End Screen Click-Through Rate (%)** – Percentage of viewers who clicked on end-screen elements.
56. **End Screen Impressions** – Number of times end-screen elements were displayed.
57. **End Screen Clicks** – Total number of clicks on end-screen elements.
58. **Teaser Click-Through Rate (%)** – Percentage of viewers who clicked on teaser elements.
59. **Teaser Impressions** – Number of times teaser elements were displayed.
60. **Teaser Clicks** – Total number of clicks on teaser elements.
61. **Card Click-Through Rate (%)** – Percentage of viewers who clicked on cards.
62. **Card Impressions** – Number of times cards were displayed.
63. **Card Clicks** – Total number of clicks on cards.

Playlist and clip metrics:

64. **Views per Playlist Start** – Average number of views per playlist start.
65. **Playlist Views** – Total number of views from playlists.
66. **Playlist Watch Time (hours)** – Total watch time from playlists in hours.
67. **Clip Watch Time (hours)** – Total watch time from clips in hours.
68. **Clip Views** – Total number of views on video clips.

Impression and click metrics:

69. **Impressions** – Number of times the video was shown to users.
70. **Video Thumbnail CTR (%)** – Click-through rate for the video thumbnail.